



Aaron Johns

AI & Software Developer

+91 8714013438

aaronjohns.dev

aaronjohns2004@gmail.com

Kuthiravattathil House,
Eravimangalam PO, Kottayam,
Kerala, 686613

ABOUT ME

AI and Data Science undergraduate with experience in drone projects, software development, and technical competitions. Interested in building intelligent systems and expanding my understanding of emerging technologies.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

St. Joseph's College of Engineering & Technology, Palai
CGPA: 7.71/10 | 2022 – 2026

De Paul Public School, CBSE, Kuravilangad, Kerala

XII: 78% | 2022 X: 89% | 2020

EXPERTISE

- Python
 - OpenCV, matplotlib, pytorch, pymavlink, pandas, numpy
- Data Analysis: SQL & R
- Git/GitHub
- VS Code
- Linux
- MavSDK
- Gazebo
- Hardware
 - Embedded Systems, Vision Sensors, RF Systems, Serial Interfaces

LANGUAGES

- English (Fluent)
- Malayalam (Fluent)
- Arabic (Reading)

EXPERIENCE

College Achievements

Courses & Leadership

- NIELIT Calicut – Allied Technologies for UAS/Drone – Participation (Explored UAV systems, payload design, and communication protocols)
- Founder, Drone Club – SJCT: Established the college's first drone club.
- NIDAR 2025 Software Head: Headed the software division to develop an autonomous drone system.
- Vitals 24 Team Lead: Led the team to build "Bromance," an ADHD-focused accessibility app.

Competitions & Projects

Competitions

- NIDAR 2025 – Overall AIR 12 | Design Review AIR 5 | Business Plan AIR 8 | Ignite Prize AIR 13
 - Developed an autonomous drone system tailored for precision agriculture, optimizing payload drop mechanisms and custom imaging workflows for efficient crop monitoring
- SAEIndia Autonomous Drone Development Challenge 2026 – AIR 20
 - Developed QR detection algorithm and automated payload drop mechanism for a fully autonomous drone in live flight competition.
- SAEIndia Autonomous Drone Development Challenge 2025 – AIR 17 & Kerala 2nd
- IEEE, Sahridaya College – Vitals 24 – 1st Place – Prize \$500 USD
 - Created app for ADHD students; outperformed 700+ teams
- Sacred Hearts College, Kochi – Blitzkrieg Hackathon – 1st Place – Prize INR 8,000
 - Solved 4 timed algorithmic challenges under a 3-hour constraint; outperformed 10+ teams.
- Sacred Hearts College, Kochi – Astra 2024 Secret Syntax – 1st Place – Prize INR 4,000
 - Wrote code without help of a screen.
- ADSA, SJCT – Codezilla Hackathon – 2nd Place
 - Completed multiple timed coding challenges focused on optimization and logic.

Projects

- Long-Context token Gater
 - Developed a modular system to optimize how LLMs process long-form data.
 - Implemented specialized gating modules using hybrid, joint entropy, and quantum-based logic.
- AI-Driven Pandemic Simulation (GAMA Platform) – Main Project
 - Implemented SEIDR and multi-fidelity physics-informed neural networks
 - Engineered agent decision rules and physics-based constraints for high-fidelity infection modeling.
 - Developed real-time visualizations and comparative analytics for policy impact assessment.
- Bromance App: ADHD Attention Methods
 - Interactive Querying: Replaced passive reading with an active retrieval system